
BattingEdge – Comprehensive Project Audit Report

Date: 23 November 2025
Environment: Python 3.11.9
Working Directory: d:\Users\Anoshia\BattingEdge_FYP\notebooks

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1. Verified Python Environment & Installed Libraries

All core ML, CV, and backend libraries were successfully detected:

- TensorFlow 2.19.1
- Keras 3.12.0
- Mediapipe 0.10.14
- OpenCV 4.12.0
- NumPy 2.1.3
- Pandas 2.3.3
- Scikit-learn 1.7.2
- FastAPI 0.121.1

The environment is stable and ready for both training and inference workflows.

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2. Jupyter Notebook Inventory (Consolidated)

A total of **26 notebooks** were discovered. Key functional categories include:

Diagnostics & Debugging

- 0_Debug_Paths.ipynb
- 0_Debug_Shape.ipynb
- 0_Debug_V7_Crash.ipynb

Utilities & Dataset Processing

- CSV generation utilities
- Video counting and dataset organization scripts

- Feature verification and unification utilities

Feature Engineering

- 2D_Feature_Engineering_HF_Local.ipynb
- 2E_Feature_Engineering_V7.ipynb
- 2F_Feature_Engineering_V7_Clean.ipynb

Model Training (All Versions)

Includes successful and failed experiments for transparency:

- V3 Random Forest (failed)
- V4 LSTM (failed)
- V5 Fixed + Shot Classifier
- V6 FineTune (failed)
- V7 Diagnostics
- V7 Shot Classifier
- **V8 Final Shot Classifier (Active Model)**

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3. Saved Models Summary

A total of **24 saved model artifacts** were found. Important models:

Primary Shot Classification Models

- shot_model_V8_best.keras
- shot_model_V8_final.keras
- shot_brain_V8.keras

Legacy & Support Models

- shot_model_V7.keras
- base_model_V7_BRAIN.keras
- Encoder & scaler files for V7/V8 versions (pkl files)

All critical assets for training, fine-tuning, and inference are present.

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4. Dataset Summary

Dataset Structure

Location: data/dataset_v7_clean/

Breakdown:

- **Train:**
  - Cut: 316 videos
  - Drive: 381
  - Pull: 312
  - Sweep: 140
- **Validation:**
  - Cut: 35
  - Drive: 45
  - Pull: 47
  - Sweep: 23
- **Test:**
  - Cut: 60
  - Drive: 73
  - Pull: 60
  - Sweep: 25

Total Videos: 1,521
Total Size: 2.43 GB

The dataset is comprehensive but structurally cluttered. Folder consistency and naming standards need refinement.

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5. Extracted Features

Feature directory:
data/features/dataset_v7_99feat/

Total feature files identified: **1606**, aligned with video count.

Each feature file contains the **99-feature engineered representation** used for the LSTM-based shot classifier.

This confirms the feature generation pipeline is functioning and complete.

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6. Model Loading & Sanity Tests

Model Loading Test

Model tested: shot_model_V8_best.keras

Result:

- Loaded successfully
- Input shape: (None, 50, 99)
- Output shape: (None, 4)
- Sample prediction successful
- Example output:
    - [0.991, 0.005, 0.003, 0.000026] → Predicted: **Drive (99%)**

Mediapipe Pose Detection Test

Video: cut_test_001.avi

- Pose detected in **16/20 frames**
- Mediapipe pipeline: **operational**

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7. Project Health Summary

Operational

- Python environment
- All core libraries
- Mediapipe pipeline

- Feature extraction pipeline
- Model loading & prediction
- All key notebooks and models present
- End-to-end V8 shot classifier functioning

Needs Attention

- Dataset folder organization is messy
- Too many notebook versions (risk of confusion)
- Several deprecated test files inside venv clutter directory structure
- Feature directory contains unused remnants from old experiments
- No consolidated documentation for training → deployment workflow

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8. Recommended Immediate Actions

1. **Clean & organize notebooks**
   Reduce 26 notebooks to 6–8 core files.
2. **Archive old model versions**
   Move outdated versions to an /archive folder.
3. **Create a unified dataset check script**
   Ensures no misnamed or misplaced videos.
4. **Document the official pipeline**
   Structure should include:
   - Pose extraction
   - Feature engineering
   - Model training
   - Inference
5. **Version-lock your environment** using a requirements.txt snapshot.
6. **Prepare for production**
   Ensure inference pipeline is lightweight and clean.

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9. Most Critical Files for Future Work

Key Notebooks

- 0_Utility_Generate_Master_CSV_Clean.ipynb
- 2F_Feature_Engineering_V7_Clean.ipynb
- 3_Model_Training_V8_Shot_Classifier.ipynb

Key Models

- shot_brain_V8.keras
- base_model_V7_BRAIN.keras
- shot_encoder_V7.pkl

These are the backbone of feature extraction + prediction.

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10. Audit Completion

Full system audit successful.
All essential assets verified.
Project is functional and ready for refinement.

NOTE: It is before *inference.py* phase start.
